

Matthew Shumway

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Education

PhD, Computing and Data Sciences

Boston University, Faculty of Computing and Data Sciences

Fall 2025 – Present

Boston, Massachusetts

BS, Applied & Computational Mathematics Emphasis (ACME)

Brigham Young University

April 2025

Provo, Utah

- GPA: 4.0; Full Academic Scholarship, Dean's List
- Concentration in Data Science and Machine Learning

Publications

M. Shumway*, Y. Tu*, X. Zhang. *From Ranking to Selection: Evaluating Representation Proxies in RL*. In submission, 2026. (*equal contribution, listed alphabetically)

J. Breen, M. Kempton, A. Knudson, **M. Shumway** (α - β). *On defining Kemeny's constant for non-backtracking random walks*. Linear Algebra and its Applications, 2026. arXiv:2510.06650

Research Experience

Graduate Research Assistant

Boston University, Faculty of Computing and Data Sciences

Fall 2025 – Present

Boston, Massachusetts

- Lightweight proxies for model selection in RL: evaluated a large suite of proxies on their ability to rank and select models; proposed a new metric aligned with selection, whereas prior work evaluated only rank correlation

Spectral Graph Theory Research Assistant

Brigham Young University, Department of Mathematics

December 2023 – April 2025

Provo, Utah

- Formulated two novel analogues of Kemeny's constant for non-backtracking random walks; developed and released a documented Python package (NBRW) supporting reproducible analyses

Industry Experience

AI Engineer Intern

Vivint

May – August 2025

Lehi, Utah

- Curated personalized highlight videos from home security camera footage using vision-language models

R&D Data Science Intern

bioMérieux

July – September 2024

Salt Lake City, Utah

- Trained XGBoost models to predict the impact of DNA degeneracy on PCR-based diagnostic assay performance

Teaching

Teaching Assistant, Introduction to Probability

Boston University, Faculty of Computing and Data Sciences

Fall 2026 – Present

Boston, Massachusetts

Teaching Assistant and Course Developer, Computer Science

Brigham Young University, Department of Computer Science

January – June 2023

Provo, Utah

Awards

CDS Duan Family Fellowship, Boston University

2025

Edwin S. Hinckley Scholarship, Brigham Young University

2024